Yufei Shi

Tel/WhatsApp: (65) 87903845 | WeChat: (+86) 13363441972 | yufei005@e.ntu.edu.sg
Personal Homepage: https://yufei2000.github.io/

EDUCATION

Nanyang Technological University (QS 12) **Aug 2024 – Jun 2028**
Ph.D. Candidate in Data Science

National University of Singapore (QS 8) **Aug 2022 – Jun 2023**
M.Sc. in Computer Engineering

Sichuan University (985) **Sep 2018 – Jun 2022**
B.Eng. in Electronic Information Engineering
Ranking: 6/125 (4.8%); **Average Score:** 88.7 ; **Honors:** National Inspirational Scholarship; Sanxin Scholarship (Top 3%); Outstanding Graduate of Sichuan University; MCM/ICM Finalist.

PROJECT EXPERIENCE

One Sentence, One Drama **Under Review, 2026 | First Author**
Keywords: LLM Agents, Personalized Generation, Long Video Generation, Text-to-Video
- Built a hierarchical multi-agent system that turns one sentence into plot beats, scenes, characters, image/video prompts, audio cues, and transitions.
- Designed debate-based story planning to sharpen opening hooks, conflict escalation, pacing, and cliffhangers.
- Used 3D world priors, character representations, and 360-degree panoramas to keep character positions and scene layouts consistent across clips.
- Built review loops for scripts, first frames, video, and audio. Matched music and planned transitions to improve continuity.

PVChat: Personalized Video Chat with One-Shot Learning **ICCV 2025 | First Author**
Keywords: Personalized Video Understanding
- Proposed PVChat, a Video LLM that recognizes and reasons about a person from one reference video.
- Automated video-QA augmentation with identity-preserving positives, hard negatives, and questions on existence, appearance, action, and location.
- Designed ReLU-routed Mixture-of-Heads with Smooth Proximity Regularization and Head Activation Enhancement to reduce identity confusion.
- Trained progressively from static identity cues to dynamic video across healthcare, daily-life, anime, and real-world footage.

PVChat-R1: Reinforced Personalized Video Chat with One-Shot Learning **Under Review, 2026 | First Author**
Keywords: Personalized Video Understanding, Reinforcement Learning, GSPO
- Extended PVChat with CA-Dynamic-GSPO, using hierarchical rewards for response validity, identity correctness, and content fidelity.
- Adapted candidate group sizes and strengthened ground-truth supervision when no candidate qualified. Expanded the corpus to 73 subjects and added minute-long videos.

AV-GRPO: Joint Audio-Video Generation **Under Review, 2026 | Main Contributor**
Keywords: Diffusion Reinforcement Learning, Audio-Video Generation, Synchronization
- Contributed to online diffusion RL that anchors one modality during rollouts and optimizes the other via trajectory locking for clearer reward attribution.
- The study introduced difficulty-controlled 5DAV. On JavisBench and VABench, AV-GRPO surpassed LTX-2.3 in quality, semantic alignment, and synchronization.

SciEducator: Scientific Video Understanding and Educating **CVPR 2026 | Major Contributor**
Keywords: Multimodal LLM, Video Agent, RAG, Tool-use, Scientific Video QA, Educational Content Generation
- Guided the design of a Deming-cycle agent system for scientific video understanding, combining planning, evaluation, retrieval, captioning, and tool use.
- Designed a pipeline that uses retrieval and tools to turn video phenomena into children's e-booklets with procedures, safety tips, illustrations, and narration.
- Helped evaluate video QA and teaching content on SciVBench, where SciEducator outperformed GPT-4o, Gemini, Claude, and prior video agents.

4DPC^{2}hat: Dynamic Point Cloud Understanding **ICML 2026 | Major Contributor**

Keywords: 4D Point Cloud MLLM, Spatial-Temporal Reasoning, Mamba, Point-BERT, Failure-Aware Bootstrapping

- Co-designed a dynamic point-cloud MLLM using Point-BERT, bidirectional Mamba, and LLaMA to model local geometry and long-range motion for 4D QA.
- Helped build 4DPC^{2}hat-200K with topology-consistent sampling, two-level captions, and diverse spatiotemporal QA.
- Developed failure-aware bootstrapping with targeted corrective QA, reaching a 78.01 GPT-4 score and 74.30% action accuracy.

ColonNeRF: Long-Sequence Colonoscopy Reconstruction **Neurocomputing IF=6.5 | First Author**

Keywords: NeRF, Neural Rendering, Medical Imaging, 3D Reconstruction, Camera Pose Densification

- Proposed ColonNeRF for novel-view synthesis in long colonoscopy sequences with complex folds and sparse camera trajectories.
- Designed region division and integration to reconstruct the colon in segments and reduce geometric inconsistency.
- Built multi-level fusion and DensiNet pose densification to improve geometry under sparse views.
- Validated on synthetic and real colonoscopy data, improving LPIPS-ALEX by 67%–85% over prior NeRF baselines.

Tune-A-Video **ICCV 2023 | GitHub Star 4.3k+**

Keywords: text-to-video

- Contributed to one-shot video tuning framework that adapts pretrained text-to-image diffusion models to text-to-video generation using spatio-temporal attention and DDIM-inversion-based structure guidance.
- Conducted evaluation and benchmarking against representative T2V generation/editing baselines, analyzing temporal consistency, prompt alignment, structure preservation, style-transfer quality, and typical failure cases.

WORK EXPERIENCE

Kwai **Feb 2024 – May 2024**

Senior Engineer of Machine Learning, Department of LLM and Multimodal Algorithms, Beijing

- Optimized multimodal content detection algorithms, utilizing CLIP and transformers to fuse image, text, and audio data, enhancing Kuaishou's original content protection effectiveness and user experience.

Sony China **Feb 2022 – May 2022**

Algorithm Research Intern, Center for Research and Development

- Participated in a deep learning-based diseased chicken detection project; implemented and fine-tuned SSD, FSSD, and IFSSD models for image-based disease identification, achieving 95% accuracy on the validation set.

PATENTS AND PAPERS

Yufei Shi, et al. PVChat: Personalized Video Chat with One-Shot Learning. **ICCV 2025**.

Yufei Shi, et al. PVChat-R1: Reinforced Personalized Video Chat with One-Shot Learning. **Under Review**.

Zhiyu Xu, Weilong Yan, **Yufei Shi**, et al. AV-GRPO: Modality-Anchored Decoupling Diffusion Reinforcement Learning for Joint Audio-Video Generation. arXiv:2609.29816. **Under Review**.

Yufei Shi, Beijia Lu, Jia-Wei Liu, Ming Li, Mike Zheng Shou. ColonNeRF: Hierarchical Neural Radiance Fields for High-Fidelity Long-Sequence Colon Reconstruction. **Neurocomputing IF=6.5**.

Yufei Shi, et al. One Sentence, One Drama: Personalized Short-Form Drama Generation via Multi-Agent Systems. **Under Review**.

Xindan Zhang, Weilong Yan, **Yufei Shi**, et al. 4DPC^{2}hat: Towards Dynamic Point Cloud Understanding with Failure-Aware Bootstrapping. **ICML 2026**.

Zhiyu Xu, Weilong Yan, **Yufei Shi**, et al. SciEducator: Scientific Video Understanding and Educating via Deming-Cycle Multi-Agent System. **CVPR 2026**.

Yucheng Chen, Yang Yu, **Yufei Shi**, et al. RIHA: Report-Image Hierarchical Alignment for Radiology Report Generation. **IEEE JBHI IF=6.5, SCI Q1**.

Jay Zhangjie Wu, Yixiao Ge, Xintao Wang, Stan Weixian Lei, Yuchao Gu, **Yufei Shi**, Wynne Hsu, Ying Shan, Xiaohu Qie, Mike Zheng Shou. Tune-A-Video: One-Shot Tuning of Image Diffusion Models for Text-to-Video Generation. GitHub: https://github.com/showlab/Tune-A-Video, **Star 4.3k**, **ICCV 2023**.

Utility Patent Based on Near Infrared Light Local Cerebral Oxygen Saturation Detection Equipment Patent. **First Inventor**.

Utility Patent an infrared communication device with relay and forwarding function. **First Inventor**.